

Chemistry — class test

Chemistry (4CH1) · IGCSE · total 20 marks · name: _____

Question 1 (1 marks · medium)

4CH1-S4-g

(a) 4CH1-4.41C

[1]

Propanoic acid reacts with ethanol in an esterification reaction.

What is the displayed formula of the ester that is formed during this reaction?

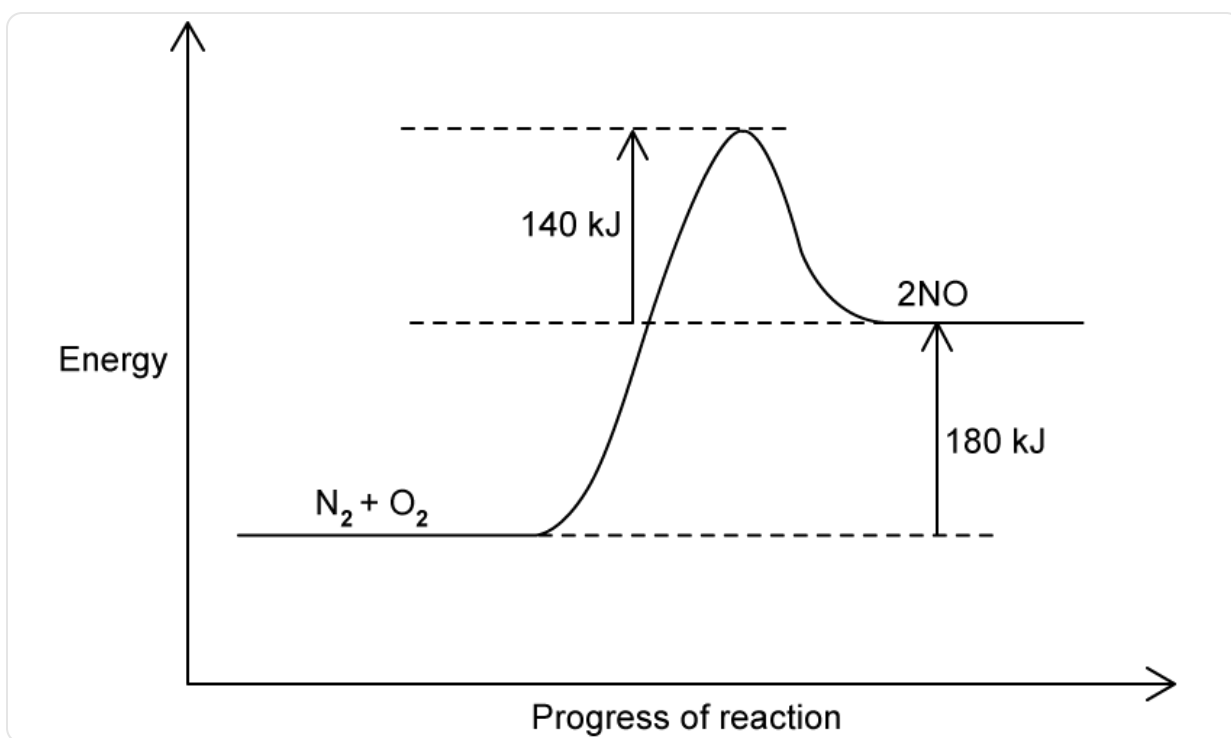
Question 2 (1 marks · easy)

4CH1-S3-a

(a) 4CH1-3.5C

[1]

The reaction profile for making nitrogen monoxide, NO, is shown below:



What is the activation energy for this reaction?

- A) +140 kJ
- B) +180 kJ
- C) +320 kJ
- D) -330 kJ

Question 3 (1 marks · easy)

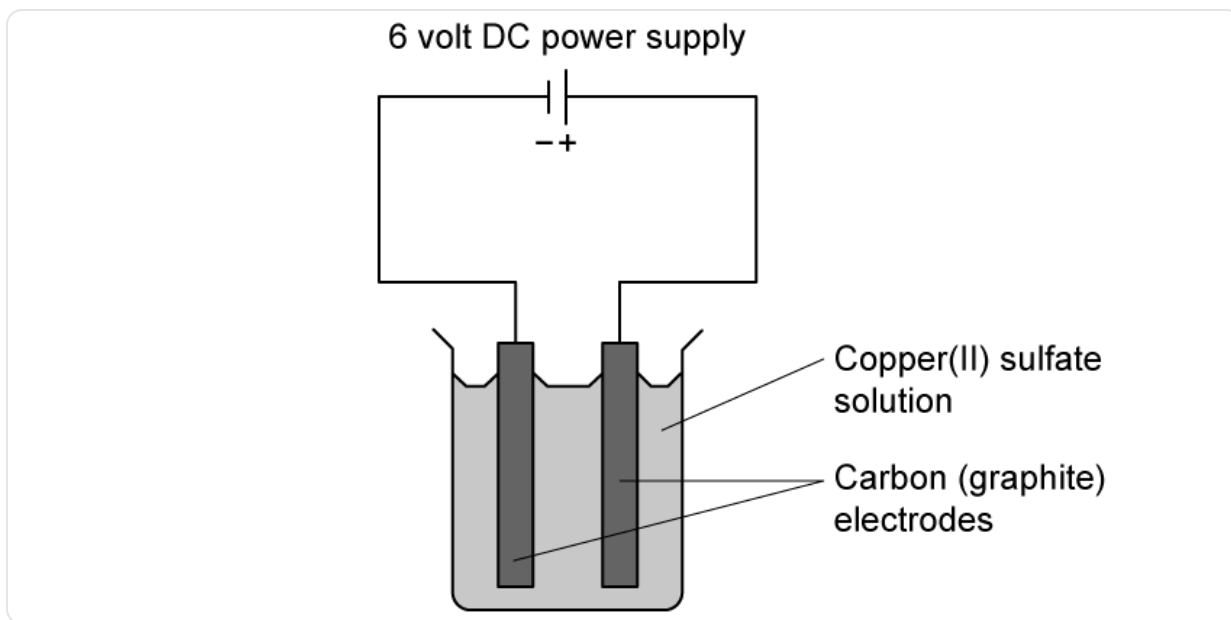
4CH1-S1-i

(a) 4CH1-1.58C, 4CH1-1.60C

[1]

Separate: Chemistry Only

A student uses the apparatus to electrolyse an aqueous solution of copper(II) sulfate.



Which substances are formed at the negative electrode (cathode) and the positive electrode (anode)?

- A) Negative electrode: hydrogen Positive electrode: sulfur
- B) Negative electrode: copper Positive electrode: oxygen
- C) Negative electrode: sulfur Positive electrode: hydrogen
- D) Negative electrode: oxygen Positive electrode: copper

Question 4 (1 marks · easy)

4CH1-S1-e

(a) 4CH1-1.26

[1]

A company produces tin from an ore called cassiterite, which is mainly tin oxide.

What is the relative formula mass (M_r) of tin oxide, SnO_2 ?

[A_r of Sn = 119; O = 16]

- A) 135
- B) 151
- C) 270
- D) 1904

Question 5 (1 marks · easy)

4CH1-S2-f

(a) 4CH1-2.28

[1]

A student is carrying out a titration using phenolphthalein indicator.

What is the correct colour of this indicator in an acid and alkali?

- A) Colour in acid: pink Colour in alkali: colourless
- B) Colour in acid: colourless Colour in alkali: pink
- C) Colour in acid: red Colour in alkali: yellow
- D) Colour in acid: orange Colour in alkali: red

Question 6 (1 marks · easy)

4CH1-S2-b

(a) 4CH1-2.5

[1]

Which element in Group 7 is a solid at room temperature?

- A) fluorine
- B) chlorine
- C) bromine
- D) iodine

Question 7 (1 marks · medium)

4CH1-S2-h

(a) 4CH1-2.46, 4CH1-2.47

[1]

A compound is suspected of containing copper ions, Cu^{2+} and bromide ions, Br^- .

Which test would **not** help to positively identify copper ions or bromide ions?

- A) Test: flame test Result: blue-green flame
- B) Test: add sodium hydroxide solution Result: blue precipitate forms
- C) Test: add acidified silver nitrate solution Result: cream precipitate forms
- D) Test: add sodium hydroxide solution Result: green precipitate forms

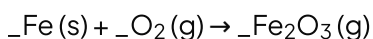
Question 8 (1 marks · easy)

4CH1-S1-e

(a) 4CH1-1.25

[1]

What numbers will balance the following equation?



- A) 2,3,4
- B) 4,3,2
- C) 2,3,1
- D) 1,2,3

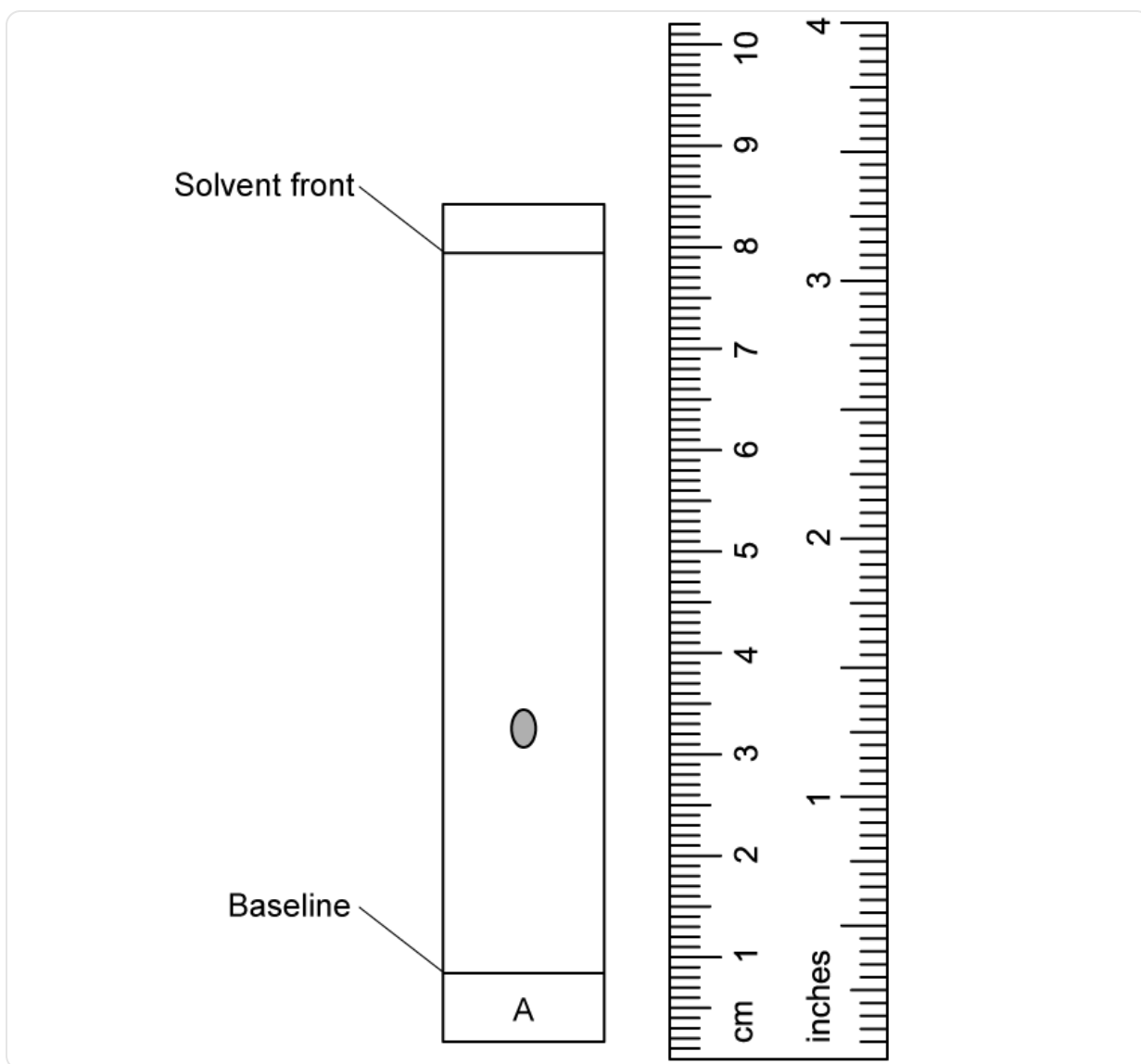
Question 9 (1 marks · medium)

4CH1-S1-b

(a) 4CH1-1.11, 4CH1-1.12

[1]

A student obtained the following chromatogram for substance A.



Determine the R_f value of substance A.

- A) 0.66
- B) 0.26
- C) 0.35
- D) 0.43

Question 10 (1 marks · easy)

4CH1-S2-d

(a) 4CH1-2.18

[1]

What type of reaction occurs when iron rusts?

- A) Oxidation
- B) Neutralisation
- C) Combustion
- D) Displacement

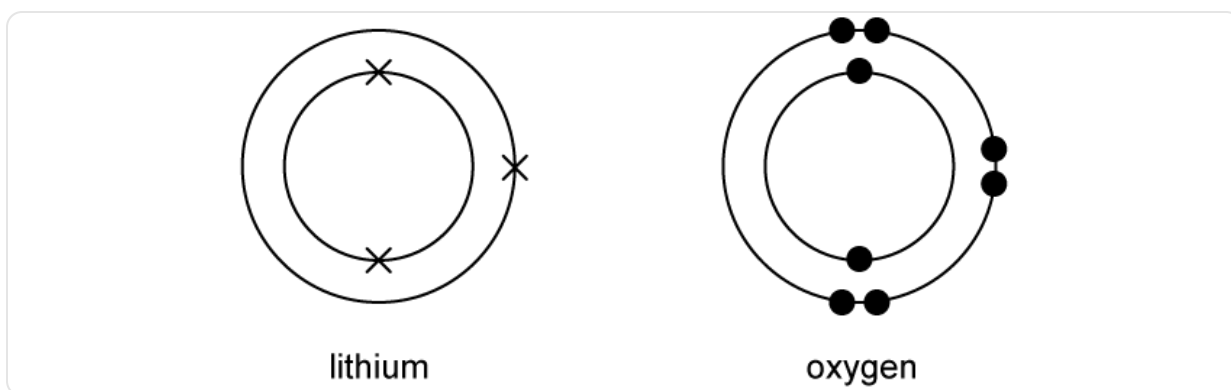
Question 11 (1 marks · medium)

4CH1-S1-f

(a) 4CH1-1.37

[1]

The diagram shows the electronic configurations of an atom of lithium and an atom of oxygen.



Which statement is **not** correct about the changes in electronic configuration that occur when lithium and oxygen react to form lithium oxide, Li_2O ?

- A) Two lithium atoms each lose one electron
- B) One oxygen atom gains 2 electrons
- C) A lithium ion has an electronic configuration of 2
- D) An oxide ion has an electronic configuration of 2.6

Question 12 (1 marks · easy)

4CH1-S2-f

(a) 4CH1-2.31

[1]

Carbonic acid is found in fizzy drinks and has the formula, H_2CO_3 .

Which ion causes the drink to be acidic?

- A) OH^-
- B) CO_3^{2-}
- C) H^+
- D) O^{2-}

Question 13 (1 marks · medium)

4CH1-S1-g

(a) 4CH1-1.46

[1]

Carbon dioxide contains covalent bonds.

The covalent bonds in a molecule of carbon dioxide can be represented by a dot-and-cross diagram.

Which is the correct dot-and-cross diagram for carbon dioxide?

Question 14 (1 marks · medium)

4CH1-S4-a

(a) 4CH1-4.6

[1]

Which of the following reactions is an example of an addition reaction?

- A) $\text{CH}_4 + \frac{1}{2}\text{O}_2 \rightarrow \text{CO} + 2\text{H}_2\text{O}$
- B) $\text{C}_2\text{H}_4 + \text{Br}_2 \rightarrow \text{C}_2\text{H}_4\text{Br}_2$
- C) $\text{CH}_4 + 2\text{O}_2 \rightarrow \text{CO}_2 + 2\text{H}_2\text{O}$
- D) $\text{CH}_4 + \text{Br}_2 \rightarrow \text{CH}_3\text{Br} + \text{HBr}$

Question 15 (1 marks · medium)

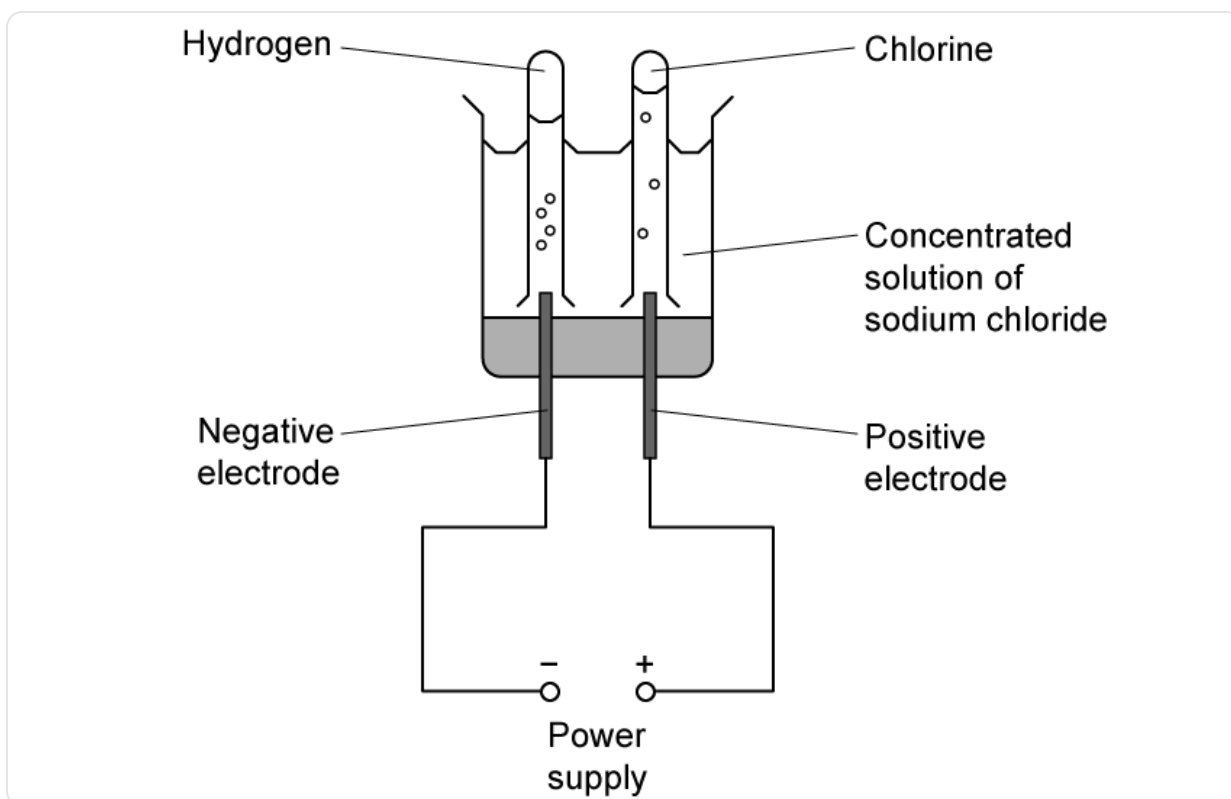
4CH1-S1-i

(a) 4CH1-1.59C

[1]

Separate: Chemistry Only

Hydrogen gas and chlorine gas can be prepared in a laboratory by electrolysis of a concentrated solution of sodium chloride.



Which is the ionic half-equation that represents the formation of chlorine gas at the positive electrode?

- A) $\text{Cl}^- \rightarrow \text{Cl} + \text{e}^-$
- B) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
- C) $\text{Cl}_2 + 2\text{e}^- \rightarrow 2\text{Cl}^-$
- D) $2\text{Cl}^- + 2\text{e}^- \rightarrow \text{Cl}_2$

Question 16 (1 marks · medium)

4CH1-S1-i

(a) 4CH1-1.59C

[1]

Electrolysis can be used to separate molten lead bromide.

Which statement is true for what happens at the negative electrode?

☐	A	Lead atoms are oxidised
☐	B	Lead atoms are reduced
☐	C	Lead ions are oxidised
☐	D	Lead ions are reduced

- A) Lead atoms are oxidized
- B) Lead atoms are reduced
- C) Lead ions are oxidized
- D) Lead ions are reduced

Question 17 (1 marks · medium)

4CH1-S4-e

(a) 4CH1-4.31C, 4CH1-4.33C

[1]

Which of the following reactions does **not** involve the oxidation of ethanol?

- A) Production of ethanol from the fermentation of glucose
- B) Heating ethanol with potassium dichromate(VI) in dilute sulfuric acid
- C) Reaction of ethanol with oxygen in the air
- D) Formation of carbon dioxide and water from the complete combustion of ethanol

Question 18 (1 marks · medium)

4CH1-S2-g

(a) 4CH1-2.37

[1]

Which equation does **not** show the correct reaction of an acid?

- A) copper oxide + hydrochloric acid → copper chloride + water
- B) calcium carbonate + nitric acid → calcium nitrate + carbon dioxide
- C) potassium hydroxide + sulfuric acid → potassium sulfate + water
- D) zinc + sulfuric acid → zinc sulfate + hydrogen

Question 19 (1 marks · easy)

4CH1-S4-b

(a) 4CH1-4.8

[1]

Fractional distillation is the process by which crude oil is separated into simpler, more useful fractions.

Which of the following statements about the process of fractional distillation is correct?

- A) The fractionating column is cooler at the bottom than at the top
- B) The fractions collected at the top of the fractionating column have a high boiling point
- C) The molecules in each fraction have a similar boiling point and carbon chain length
- D) The fractions collected at the bottom of the fractionating column have a low viscosity

Question 20 (1 marks · easy)

4CH1-S4-f

(a) 4CH1-4.35C

[1]

Ethanoic acid is a member of the homologous series of carboxylic acids.

Which is the functional group of carboxylic acids?

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